MY SQL

ASSIGNMENT 2--QUERYING DATA

```
USE employee;

INSERT INTO department (department _id, department _name) VALUES
(1, 'Software Development'),
(2, 'Marketing'),
(3, 'Data Science'),
(4, 'Human Resources'),
(5, 'Product Management'),
(6, 'Content Creation'),
(7, 'Finance'),
(8, 'Design'),
(9, 'Research and Development'),
(10, 'Customer Support'),
(11, 'Business Development'),
(12, 'IT'),
(13, 'Operations');

INSERT INTO location (Location _name) VALUES
('Chennai'),
('Bangalore'),
('Hyderabad'),
('Pune');

ALTER TABLE employees
 ADD designation VARCHAR (100),
 ADD salary DECIMAL (10,2);
```

INSERT INTO employees (employee _id, employee _name, gender, age, hire _date, designation, department _id, location _id, salary) VALUES

(5001, 'Vihaan Singh', 'M', 27, '2015-01-20', 'Data Analyst', 3, 4, 60000),

(5002, 'Reyansh Singh', 'M', 31, '2015-03-10', 'Network Engineer', 12, 1, 80000),

(5003, 'Aaradhya Iyer', 'F', 26, '2015-05-20', 'Customer Support Executive', 10, 2, 45000),

(5004, 'Kiara Malhotra', 'F', 29, '2015-07-05', NULL, 8, 3, 70000),

(5005, 'Anvi Chaudhary', 'F', 25, '2015-09-11', 'Business Development Executive', 11, 1, 55000),

(5006, 'Dhruv Shetty', 'M', 28, '2015-11-20', 'UI Developer', 8, 2, 65000),

(5007, 'Anushka Singh', 'F', 32, '2016-01-15', 'Marketing Manager', 2, 3, 90000),

(5008, 'Diya Jha', 'F', 27, '2016-03-05', 'Graphic Designer', 8, 4, 70000),

(5009, 'Kiaan Desai', 'M', 30, '2016-05-20', 'Sales Executive', 11, 3, 55000),

(5010, 'Atharv Yadav', 'M', 29, '2016-07-10', 'Systems Administrator', 12, 4, 80000),

(5011, 'Saanvi Patel', 'F', 28, '2016-09-20', 'Marketing Analyst', 2, 1, 60000),

(5012, 'Myra Verma', 'F', 26, '2016-11-05', 'Operations Manager', 13, 2, 95000),

(5013, 'Arnav Rao', 'M', 33, '2017-01-20', 'Customer Success Manager', 10, 3, 75000),

(5014, 'Vihaan Mohan', 'M', 30, '2017-03-10', 'Supply Chain Analyst', 10, 2, 60000),

(5015, 'Ishaan Kumar', 'M', 27, '2017-05-20', 'Financial Analyst', 7, 1, 85000),

(5016, 'Zoya Khan', 'F', 31, '2017-07-05', 'Legal Counsel', 4, 4, 100000),

(5017, 'Kabir Nair', 'M', 28, '2017-09-11', 'IT Support Specialist', 12, 2, 80000),

(5018, 'Ishan Mishra', 'M', 25, '2017-11-20', 'Research Scientist', 9, 3, 75000),

(5019, 'Ishika Patel', 'F', 29, '2018-01-15', 'Talent Acquisition Specialist', 4, 4, 55000),

(5020, 'Aarav Nair', 'M', 32, '2018-03-05', 'Software Engineer', 1, 1, 90000),

(5021, 'Advik Kapoor', 'M', 26, '2018-05-20', 'Finance Analyst', 7, 3, 85000),

(5022, 'Aadhya Iyengar', 'F', 28, '2018-07-10', 'HR Specialist', 4, 4, 60000),

(5023, 'Anika Paul', 'F', 30, '2018-09-20', 'Public Relations Specialist', 2, 2, 70000),

(5024, 'Aryan Shetty', 'M', 27, '2018-11-05', 'Product Manager', 5, 1, 95000),

(5025, 'Avni Iyengar', 'F', 31, '2019-01-20', 'Data Scientist', 3, 4, 100000),

(5026, 'Vivaan Singh', 'M', 29, '2019-03-10', 'Business Analyst', 3, 2, 75000),

(5027, 'Ananya Paul', 'F', 32, '2019-05-20', 'Content Writer', 6, 3, 60000),

(5028, 'Anaya Kapoor', 'F', 26, '2019-07-05', 'Event Coordinator', 6, 1, 60000),

(5029, 'Arjun Kumar', 'M', 33, '2019-09-11', 'Quality Assurance Analyst', 12, 2, 80000),

(5030, 'Sara Iyer', 'F', 28, '2019-11-20', 'Project Manager', 5, 1, 90000);

-- 1. Distinct Values:

--#a query to retrieve distinct salaries from the Employees table.

```
SELECT DISTINCT salary FROM employees;
```

-- 2 Alias (AS):

--# Provide aliases for the "age" and "salary" columns as "Employee _Age" and Employee _ Salary, respectively.

```
SELECT age AS 'Employee _Age', salary AS 'Employee _Salary' FROM employees;
```

#3. Where Clause & Operators:

--#Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

```
SELECT * FROM employees WHERE salary > 50000 AND hire _date <'2016-01-01';
```

--#Find the employee whose designation is missing and fill it with "Data Scientist".

```
SELECT * FROM employees WHERE designation IS NULL;
UPDATE employees SET designation = 'Data Scientist' WHERE designation IS NULL;
```

--#Sorting and Grouping Data:

--#4 ORDER BY:

-- Find employees sorted by department ID in ascending order and salary in descending order.

```
SELECT * FROM employees ORDER BY department _id ASC, salary DESC;
```

--#5 LIMIT:

--#Display the first 5 employees hired in the year 2018.

```
SELECT * FROM employees WHERE hire _date >= '2018-01-01'
AND hire _date < '2019-01-01'
ORDER BY hire _date ASC
LIMIT 5;
```

--#Aggregate Functions:

--#6 Calculate the sum of all salaries in the Finance department.

#--(department _id for finance department is 7)

```
SELECT department _id, SUM (salary) AS Total _Salary FROM employees
GROUP BY department _id
HAVING department _id = '7';
```

--#7 Find the minimum age among all employees.

```
SELECT MIN (age) AS minimum _age FROM employees;
```

--#GROUP BY:

#--8 List the maximum salary for each location.

```
SELECT location _id, MAX (salary) AS max _salary FROM employees
GROUP BY location _id;
```

#--9 -Calculate the average salary for each designation containing the word 'Analyst'.

```
SELECT designation,
 AVG (salary) AS average _salary
 FROM employees
 GROUP BY
 DESIGNATION
 HAVING designation like '%analyst%';
```

##HAVING:

--#10 Find departments with less than 3 employees.

```
SELECT department _id, count (*) AS employee _count
FROM employees
GROUP BY department _id
HAVING COUNT (*) < 3;
```

--#11 Find locations with female employees whose average age is below 30.

```
SELECT location _id,
AVG (age) AS Average _age
FROM employees
WHERE gender ='F'
GROUP BY location _id
HAVING AVG (age) < 30;
```

#--Joins:

--#Inner Join:

--#12 List employee names, their designations, and department names where employees are assigned to a department.

```
SELECT
e. employee _name, e. designation, d. department _name
FROM employees e
INNER JOIN
department d
ON e. department _id = d. department _id;
```

#-- Left Join:

#-13 List all departments along with the total number of employees in each department, including departments with no employees.

```
SELECT
d. department _id, d. department _name,
COUNT (e. employee _id) AS total _employees
FROM department d
LEFT JOIN employees e
ON d. department _id = e. department _id
GROUP BY
d. department _id,
d. department _name;
```

#--14 Right Join:

--# Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

```
SELECT
l. location _name, e. employee _name
FROM employees e
RIGHT JOIN location l
ON e. location _id = l. location _id;
```